FLUOXETINE

Suicidality <25 yr — monitor

Long half-life

CHEMICAL NAME (RS)-N-methyl-3-phenyl-3-[4-(trifluoromethyl)phenoxy]propan-1-amine (hydrochloride)

CLASS Selective serotonin reuptake inhibitor (SSRI)

TRADE NAMES Prozac · Flutine · Sarafem · Prozep

THERAPEUTIC USES / INDICATIONS

- Major depression (first-line in children/adolescents).
- Obsessive-compulsive disorder; bulimia nervosa.
- Panic disorder; premenstrual dysphoric disorder.

ADMINISTRATION

- **PO** once daily in the morning (may cause insomnia), with or without food.
- Benefit takes 2–4 weeks; continue ≥6 months after remission.

DOSAGES

- **Depression:** 20 mg once daily; may increase to 60 mg.
- **OCD:** 20–60 mg daily.
- **Bulimia:** 60 mg daily.
- **Children 8–18 yr (depression):** 10 mg daily, up to 20 mg.
- Elderly: max 40 mg.

PHARMACOKINETICS

- **Onset:** 2–4 weeks (antidepressant). **Peak:** 6–8 h.
- Metabolised by CYP2D6 to active norfluoxetine; strong CYP2D6 inhibitor.
- **t½:** 1–4 days (norfluoxetine 7–15 days) → low withdrawal risk; wash-out 5 weeks before MAOI.

PHARMACODYNAMICS

- Inhibits **serotonin (5-HT) reuptake** at presynaptic neurons → ↑ synaptic serotonin → antidepressant, anti-obsessional effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, diarrhoea, headache.
- Insomnia, anxiety, agitation (early).
- Sexual dysfunction; sweating.

ADVERSE EFFECTS (SERIOUS)

- Suicidal thoughts/behaviour in young people (<25 yr) early in treatment.
- Serotonin syndrome; hyponatraemia (SIADH, elderly).
- GI bleeding (with NSAIDs/anticoagulants); QT prolongation.
- Mania activation; seizures.

CONTRAINDICATIONS & CAUTIONS

- MAOIs (within 14 days before; 5 weeks after fluoxetine).
- Manic phase; **caution:** epilepsy, bleeding disorders, bipolar disorder.

KEY INTERACTIONS

- **MAOIs, linezolid, tramadol, triptans, St John's wort:** serotonin syndrome.
- **CYP2D6 substrates (TCAs, metoprolol, tamoxifen, codeine):** ↑ levels / ↓ activation.
- **NSAIDs, aspirin, warfarin:** ↑ bleeding.

NURSING MANAGEMENT

- Assess mood and suicide risk, especially first weeks and dose changes.
- Monitor Na^+ in elderly.
- Watch for agitation, mania, serotonin toxicity.
- Support adherence.

PATIENT EDUCATION

- Benefit takes 2–4 weeks; keep taking.
- Report worsening mood, thoughts of self-harm, agitation immediately.
- Take in the morning.
- Do not stop suddenly (though withdrawal is milder than with other SSRIs).

KEY POINTS

- SSRI with very long half-life.
- Suicidality warning <25 yr.
- 20 mg daily; first-line in young people.
- Strong CYP2D6 inhibitor.